5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.

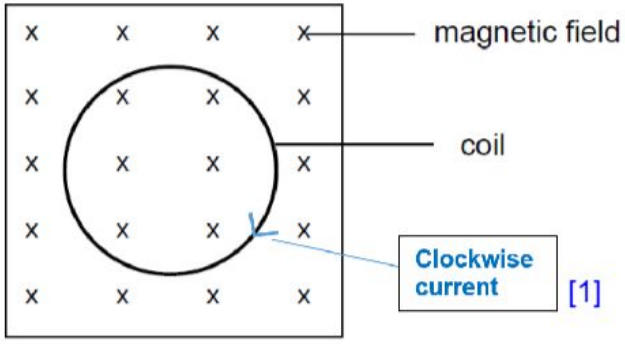


Fig. 5.1

(a) Define *magnetic flux*.

The magnetic flux through a surface is the product of the component of the magnetic flux density normal to the surface and the area of the surface [1].

[1]

(b) (i) The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

Since the flux density B decreases, then flux linkage $\Phi = BA$ decreases. [1]

By Faraday's Law, there is an induced emf in the coil. Since the coil is a closed circuit, the induced emf produces an induced current. [1] [2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

By Lenz's Law, the direction of the induced current must be such as to produce a magnetic flux into the page to *oppose* the decrease of the magnetic flux linkage. [1] Therefore, by the Right Hand Grip Rule, the current will be in the *clockwise* direction to produce an induced magnetic field in the same direction. [2]

[1 for correct direction of induced current]

(c) A magnet of length x and magnetic flux density B is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height h above the top of a *long* solenoid of length L and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet x is much smaller than the length of the solenoid L.

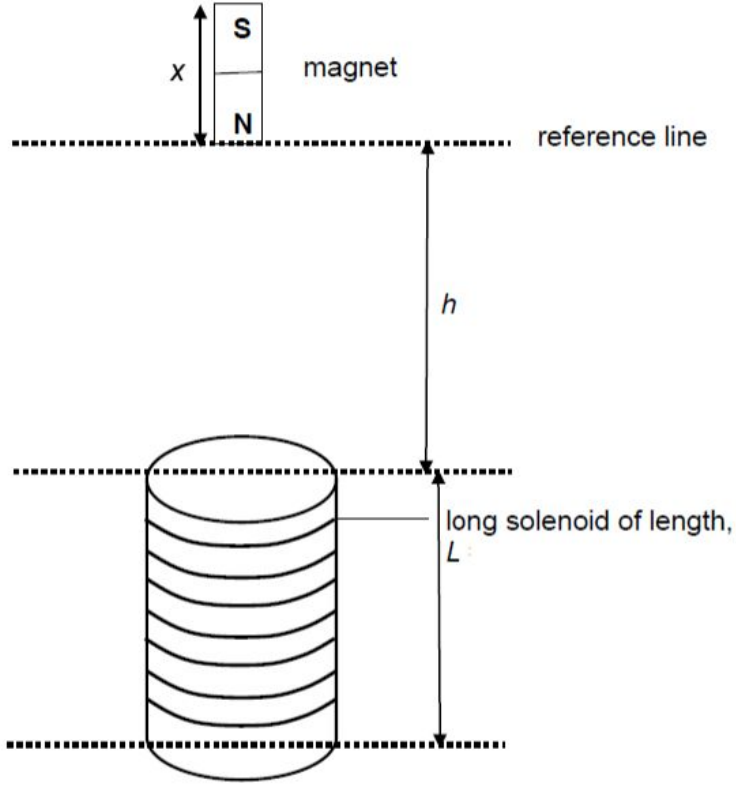


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)

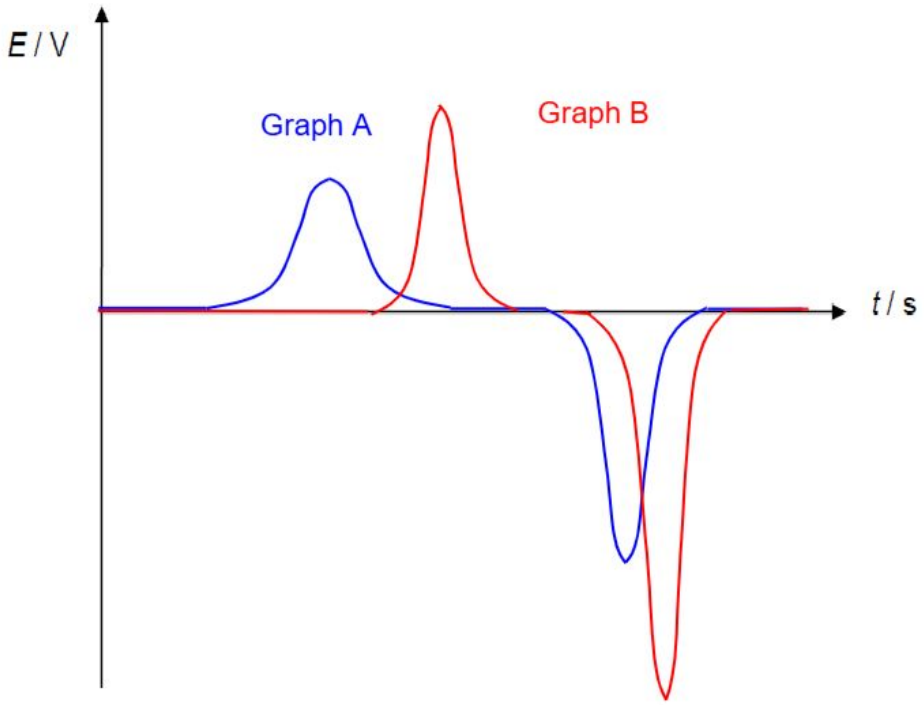


Fig. 5.3

[2]

(ii) State how your graph in part **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

The new graph will look like a reflection of Graph A about the time axis [1].

.....

.....[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

Both peaks are higher and narrower [1]

Both peaks occur at later time [1]

[2]

[Total: 10]